

Exchangeable Dyads: Model Specification and Back-Transformation

Pascal Küng

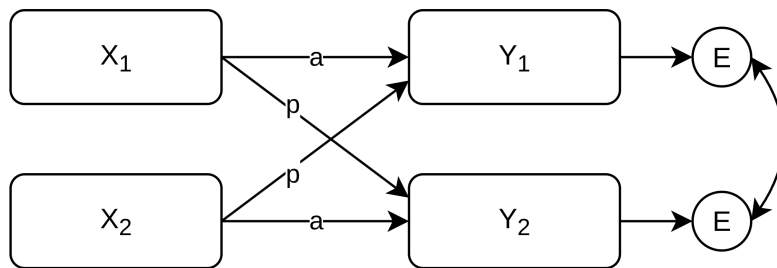
Full GitHub repository: The full repository contains the tutorial source files, reusable helper functions, model code, rendered PDFs, and additional information: <https://github.com/Pascal-Kueng/05DyadicDataAnalysis>

1 Background

Exchangeable dyads do not have substantively meaningful partner roles. Partner labels are arbitrary, and flipping the labels should not change the model interpretation.

The full intensive-longitudinal specification used here models (Bolger & Laurenceau, 2013; del Rosario & West, 2025):

- pooled fixed effects, because the two partner labels are not substantively different;
- pooled average time trends and pooled actor/partner effects;
- stable between-dyad differences in average levels and slopes;
- partner deviations from those dyad-level effects, because the two people in a dyad are still distinct repeated-observation units;
- correlations among the dyad-level random effects and among the partner deviation random effects;
- same-day residual interdependence between partners after fixed and random effects have been accounted for;
- a residual structure that remains invariant when the arbitrary partner labels are switched.



When both are requested, `prepare_dyad_data()` creates the APIM and DIM predictors together. Omitting `role` states the exchangeability assumption, while `seed` makes its arbitrary member contrast reproducible.

2 Data structure

The 12 rows below only illustrate the prepared structure; the model specifications assume study-sized data, and the output uses the cached fit to the main tutorial's 11,000-row simulation.

```
exchangeable_raw <- tibble(
  coupleID = rep(31:32, each = 6),
  userID = rep(c("31_1", "31_2", "32_1", "32_2"), each = 3),
  diaryday = rep(0:2, 4),
  closeness = c(5.2, 5.5, 5.1, 4.8, 5.0, 5.3,
                4.9, 5.1, 5.0, 5.4, 5.2, 5.5),
  provided_support = c(4.1, 4.5, 4.3, 3.9, 4.2, 4.4,
                       4.0, 4.3, 4.1, 4.6, 4.5, 4.7)
)

df_long_dim <- prepare_dyad_data(
  data = exchangeable_raw,
  dyad = coupleID,
  member = userID,
  time = diaryday,
  predictors = provided_support,
  model_types = c("apim", "dim"),
  seed = 123
) |>
mutate(
  diaryday_c = diaryday - mean(diaryday)
)
```

Table 1: Prepared rows for an exchangeable dyad

userID	diaryday	.member_contrast_arbitrary	.provided_support_cwp_actor	.provided_support_cwp_partner
31_1	0	-1	-0.2000000	-0.2666667
31_1	1	-1	0.2000000	0.0333333
31_1	2	-1	0.0000000	0.2333333
31_2	0	1	-0.2666667	-0.2000000
31_2	1	1	0.0333333	0.2000000
31_2	2	1	0.2333333	0.0000000
32_1	0	-1	-0.1333333	0.0000000
32_1	1	-1	0.1666667	-0.1000000
32_1	2	-1	-0.0333333	0.1000000
32_2	0	1	0.0000000	-0.1333333
32_2	1	1	-0.1000000	0.1666667
32_2	2	1	0.1000000	-0.0333333

.member_contrast_arbitrary is constant within person and opposite within dyad. Its sign has no substantive meaning.

3 glmmTMB model

```
model_exch_long_apim_glmmtnb <- glmmTMB(
  closeness ~ 1 +

  # Pooled fixed effects: no gender-specific intercepts or slopes.
  diaryday_c +
  .provided_support_cwp_actor +
  .provided_support_cwp_partner +
  .provided_support_cbp_actor +
  .provided_support_cbp_partner +

  # Dyad-level means and slopes.
  (1 + diaryday_c +
```

```

        .provided_support_cwp_actor +
        .provided_support_cwp_partner | coupleID) +

# Partner deviations from the dyad-level effects.
# This block is kept separate from the dyad-level block so that the
# common/couple-mean and deviation components are independent.
(0 + .member_contrast_arbitrary +
  I(.member_contrast_arbitrary * diaryday_c) +
  I(.member_contrast_arbitrary * .provided_support_cwp_actor) +
  I(.member_contrast_arbitrary * .provided_support_cwp_partner) | coupleID) +

# Same-day residual interdependence:
# Independent common and member-deviation components imply homogeneous
# residual variance and covariance.
(1 | coupleID:diaryday) +
  (0 + .member_contrast_arbitrary | coupleID:diaryday),

# The Gaussian residual covariance is represented by the two blocks above.
# Without dispformula = ~ 0, glmmTMB would add a second independent
# residual variance on top of the dyadic residual covariance structure.
# For non-Gaussian families, dispformula controls family-specific dispersion
# when one exists and may be ignored otherwise; do not use ~0 as this trick.
dispformula = ~ 0,
family = gaussian(),
data = df_long_dim
)

```

The common and member-deviation components are rotated to member-level quantities in the Back-transformation section.

4 brms model

```

formula_exch_long_apim <- bf(
  closeness ~ 1 +

  # Pooled fixed effects.
  diaryday_c +
  .provided_support_cwp_actor +
  .provided_support_cwp_partner +
  .provided_support_cbp_actor +
  .provided_support_cbp_partner +

  # Dyad-level means and slopes.
  (1 + diaryday_c +
    .provided_support_cwp_actor +
    .provided_support_cwp_partner | coupleID) +

  # Partner deviations from the dyad-level effects.
  (0 + .member_contrast_arbitrary +
    I(.member_contrast_arbitrary * diaryday_c) +
    I(.member_contrast_arbitrary * .provided_support_cwp_actor) +
    I(.member_contrast_arbitrary * .provided_support_cwp_partner) | coupleID) +

  # Native same-day residual correlation, allowed to be positive or negative.
  # With exactly two members per group, unstr() estimates the single pairwise
  # correlation. For triads or larger groups it estimates a fully
  # unstructured correlation matrix.
  unstr(time = .member_contrast_arbitrary, gr = coupleID:diaryday)

  # No sigma formula is specified here. Thus, sigma ~ 1 is implied,
  # meaning one homogeneous residual variance for both partners.
  # If sigma is modelled with its own formula, brms estimates the sigma
  # coefficients on the log scale, so those coefficients must be exponentiated
  # before reporting them as residual SDs.
)

priors_exch_long_apim <- c(

```

```

prior(normal(4, 2), class = "Intercept"),
prior(normal(0, 2), class = "b"),
prior(exponential(1), class = "sd"),
prior(lkj(2), class = "cor"),
prior(student_t(3, 0, 1.5), class = "sigma"),
prior(lkj(2), class = "cortime")
)

model_exch_long_apim <- brm(
  formula = formula_exch_long_apim,
  data = df_long_dim,
  family = gaussian(link = identity),
  prior = priors_exch_long_apim,
  chains = 4,
  cores = 4,
  iter = 2000,
  warmup = 1000,
  seed = 123,
  file = file.path("brms_cache", "model_apim_ind_long_apim")
)

```

The cached illustrative fit has no HMC warnings, but a few parameters, including the intercept, mix slowly (maximum $\widehat{R} = 1.049$); refit longer before substantive inference.

5 Back-transformation

The model estimates a common/couple-mean block and a member-deviation block. For exchangeable dyads, the full APIM random-effects matrix is recovered by rotating these two blocks:

$$\text{Var}(A) = \text{Var}(B) = \Sigma_{\text{common}} + \Sigma_{\text{deviation}}$$

$$\text{Cov}(A, B) = \Sigma_{\text{common}} - \Sigma_{\text{deviation}}$$

This exact rotation assumes that the common/couple-mean and member-deviation random-effect blocks are independent.

The helper function `summarize_exchangeable_apim_brms()` used below is defined in `00_R_Functions/ReportModels.R` on GitHub. It reconstructs the common/couple-mean and member-deviation covariance matrices for every posterior draw, rotates each draw into the full APIM partner-level matrix, and then summarizes the transformed covariance, SD, correlation, fixed-effect, and residual quantities.

For `glmmTMB`, `dyadMLM::recover_exchangeable_covariance()` provides the algebraic point back-transformation. Interval estimates require a separate parametric bootstrap or transformation of asymptotic parameter draws.

```

apim_results <- summarize_exchangeable_apim_brms(
  model_exch_long_apim,
  gr = "coupleID",
  deviation_term = ".member_contrast_arbitrary",
  term_labels = c("int", "day", "actor_wp", "partner_wp"),
  partner_labels = c("A", "B")
)

```

6 Variance-covariance matrix

```
kable(
  round(apim_results$full_covariance_matrix, 3),
  caption = "Back-transformed full APIM random-effects variance-covariance matrix",
  booktabs = TRUE
) |>
  kable_styling(latex_options = "scale_down", font_size = 7)
```

Table 2: Back-transformed full APIM random-effects variance-covariance matrix

	A_int	A_day	A_actor_wp	A_partner_wp	B_int	B_day	B_actor_wp	B_partner_wp
A_int	1.180	0.011	0.096	0.018	-0.264	-0.002	-0.063	0.005
A_day	0.011	0.000	0.001	0.000	-0.002	0.000	-0.001	0.000
A_actor_wp	0.096	0.001	0.024	0.004	-0.063	-0.001	-0.018	-0.002
A_partner_wp	0.018	0.000	0.004	0.012	0.005	0.000	-0.002	0.002
B_int	-0.264	-0.002	-0.063	0.005	1.180	0.011	0.096	0.018
B_day	-0.002	0.000	-0.001	0.000	0.011	0.000	0.001	0.000
B_actor_wp	-0.063	-0.001	-0.018	-0.002	0.096	0.001	0.024	0.004
B_partner_wp	0.005	0.000	-0.002	0.002	0.018	0.000	0.004	0.012

7 SD table

```
kable(
  apim_results$sd_summary,
  digits = 3,
  caption = "Back-transformed APIM random-effect standard deviations with 95% credible intervals"
)
```

Table 3: Back-transformed APIM random-effect standard deviations with 95% credible intervals

parameter	estimate	est_error	q_low	q_high
A_actor_wp	0.153	0.018	0.119	0.191
A_day	0.018	0.001	0.016	0.020
A_int	1.085	0.056	0.987	1.206
A_partner_wp	0.107	0.020	0.066	0.144
B_actor_wp	0.153	0.018	0.119	0.191
B_day	0.018	0.001	0.016	0.020
B_int	1.085	0.056	0.987	1.206
B_partner_wp	0.107	0.020	0.066	0.144

8 Correlation matrix

```
kable(
  round(apim_results$full_correlation_matrix, 3),
  caption = "Back-transformed APIM random-effect correlation matrix",
  booktabs = TRUE
) |>
  kable_styling(latex_options = "scale_down", font_size = 7)
```

Table 4: Back-transformed APIM random-effect correlation matrix

	A_int	A_day	A_actor_wp	A_partner_wp	B_int	B_day	B_actor_wp	B_partner_wp
A_int	1.000	0.581	0.572	0.157	-0.221	-0.124	-0.377	0.043
A_day	0.581	1.000	0.369	0.146	-0.124	-0.136	-0.211	0.046
A_actor_wp	0.572	0.369	1.000	0.266	-0.377	-0.211	-0.755	-0.096
A_partner_wp	0.157	0.146	0.266	1.000	0.043	0.046	-0.096	0.230
B_int	-0.221	-0.124	-0.377	0.043	1.000	0.581	0.572	0.157
B_day	-0.124	-0.136	-0.211	0.046	0.581	1.000	0.369	0.146
B_actor_wp	-0.377	-0.211	-0.755	-0.096	0.572	0.369	1.000	0.266
B_partner_wp	0.043	0.046	-0.096	0.230	0.157	0.146	0.266	1.000

9 Final reporting summary

Table 5: Compact reporting summary

section	parameter	estimate	est_error	q_low	q_high
Fixed effects	Intercept	5.102	0.070	4.966	5.240
Fixed effects	diaryday_c	0.004	0.001	0.001	0.006
Fixed effects	.provided_support_cwp_actor	0.300	0.014	0.273	0.328
Fixed effects	.provided_support_cwp_partner	0.149	0.015	0.120	0.177
Fixed effects	.provided_support_cbp_actor	1.316	0.088	1.140	1.483
Fixed effects	.provided_support_cbp_partner	0.382	0.086	0.212	0.557
Back-transformed random-effect SDs	A_actor_wp	0.153	0.018	0.119	0.191
Back-transformed random-effect SDs	A_day	0.018	0.001	0.016	0.020
Back-transformed random-effect SDs	A_int	1.085	0.056	0.987	1.206
Back-transformed random-effect SDs	A_partner_wp	0.107	0.020	0.066	0.144
Back-transformed random-effect SDs	B_actor_wp	0.153	0.018	0.119	0.191
Back-transformed random-effect SDs	B_day	0.018	0.001	0.016	0.020
Back-transformed random-effect SDs	B_int	1.085	0.056	0.987	1.206
Back-transformed random-effect SDs	B_partner_wp	0.107	0.020	0.066	0.144
Back-transformed random-effect correlations	A_actor_wp with A_day	0.369	0.101	0.169	0.558
Back-transformed random-effect correlations	A_actor_wp with A_int	0.572	0.083	0.403	0.723
Back-transformed random-effect correlations	A_day with A_int	0.581	0.056	0.470	0.681
Back-transformed random-effect correlations	A_partner_wp with A_actor_wp	0.266	0.152	-0.028	0.558
Back-transformed random-effect correlations	A_partner_wp with A_day	0.146	0.129	-0.114	0.400
Back-transformed random-effect correlations	A_partner_wp with A_int	0.157	0.117	-0.069	0.382
Back-transformed random-effect correlations	B_actor_wp with A_actor_wp	-0.755	0.151	-0.983	-0.412
Back-transformed random-effect correlations	B_actor_wp with A_day	-0.211	0.111	-0.415	0.019
Back-transformed random-effect correlations	B_actor_wp with A_int	-0.377	0.103	-0.572	-0.163
Back-transformed random-effect correlations	B_actor_wp with A_partner_wp	-0.096	0.160	-0.412	0.215
Back-transformed random-effect correlations	B_actor_wp with B_day	0.369	0.101	0.169	0.558
Back-transformed random-effect correlations	B_actor_wp with B_int	0.572	0.083	0.403	0.723
Back-transformed random-effect correlations	B_day with A_actor_wp	-0.211	0.111	-0.415	0.019
Back-transformed random-effect correlations	B_day with A_day	-0.136	0.118	-0.362	0.094
Back-transformed random-effect correlations	B_day with A_int	-0.124	0.084	-0.288	0.040
Back-transformed random-effect correlations	B_day with A_partner_wp	0.046	0.138	-0.231	0.310
Back-transformed random-effect correlations	B_day with B_int	0.581	0.056	0.470	0.681
Back-transformed random-effect correlations	B_int with A_actor_wp	-0.377	0.103	-0.572	-0.163
Back-transformed random-effect correlations	B_int with A_day	-0.124	0.084	-0.288	0.040
Back-transformed random-effect correlations	B_int with A_int	-0.221	0.095	-0.402	-0.039
Back-transformed random-effect correlations	B_int with A_partner_wp	0.043	0.125	-0.199	0.281
Back-transformed random-effect correlations	B_partner_wp with A_actor_wp	-0.096	0.160	-0.412	0.215
Back-transformed random-effect correlations	B_partner_wp with A_day	0.046	0.138	-0.231	0.310
Back-transformed random-effect correlations	B_partner_wp with A_int	0.043	0.125	-0.199	0.281
Back-transformed random-effect correlations	B_partner_wp with A_partner_wp	0.230	0.394	-0.570	0.960
Back-transformed random-effect correlations	B_partner_wp with B_actor_wp	0.266	0.152	-0.028	0.558
Back-transformed random-effect correlations	B_partner_wp with B_day	0.146	0.129	-0.114	0.400
Back-transformed random-effect correlations	B_partner_wp with B_int	0.157	0.117	-0.069	0.382
Residual structure	residual_variance	0.878	0.012	0.854	0.903
Residual structure	same_day_residual_correlation	0.037	0.014	0.009	0.066
Residual structure	same_day_residual_covariance	0.033	0.012	0.008	0.058
Residual structure	sigma	0.937	0.007	0.924	0.950

- Bolger, N., & Laurenceau, J.-P. (2013). *Intensive longitudinal methods: An introduction to diary and experience sampling research*. Guilford Press. <https://www.guilford.com/books/Intensive-Longitudinal-Methods/Bolger-Laurenceau/9781462506781>
- del Rosario, K. S., & West, T. V. (2025). A Practical Guide to Specifying Random Effects in Longitudinal Dyadic Multilevel Modeling. *Advances in Methods and Practices in Psychological Science*, 8(3), 25152459251351286. <https://doi.org/10.1177/25152459251351286>